In-Class Activity: Thermal Expansion of FCC Al with NPT Molecular Dynamics

ASE structure generation -> LAMMPS on the SCC -> local Python analysis -> thermal expansion coefficient

The high-level problem

Materials expand when heated because atoms explore larger-amplitude, anharmonic vibrations around their equilibrium positions. In this activity, you will use molecular dynamics to measure that expansion directly. Rather than prescribing the simulation-cell size, you will allow the cell to respond to temperature while maintaining zero external pressure. From the equilibrium volume at several temperatures, you will estimate a macroscopic materials property: the coefficient of thermal expansion.

Central question: How can atomistic simulations reproduce the experimentally measurable thermal expansion of a crystalline material?

Connection to course objectives

Course idea	How this activity uses it
Choose a statistical ensemble to match the physical property.	Thermal expansion requires the simulation cell to change size, so NPT is the natural choice.
Build and transfer atomistic structures between tools.	You will create FCC Al with ASE, export it as a LAMMPS data file, and run it on the SCC.
Extract equilibrium information from fluctuating MD data.	Instantaneous volume fluctuates; the useful quantity is a time-averaged equilibrium volume.
Connect atomistic modeling to experiment.	The slope of V(T) becomes the thermal expansion coefficient, a standard experimentally measured materials property.

Overall workflow

1	ASE	Build conventional FCC Al, replicate 4 x 4 x 4, save a LAMMPS data file.
2	SCC + LAMMPS	Run five independent NPT simulations at different temperatures and zero pressure.
3	Download data	Copy the five small thermal_.dat files to your laptop.
4	Python analysis	Plot average volume vs temperature, fit V(T), and extract thermal expansion.

Part 1 - Build the FCC Al structure with ASE

Open 01_build_al_fcc_template.py on your laptop. The file already contains the imports, lattice constant, replication target, checks, and output filename. Your task is to complete only the three marked TODO sections.

1. Build conventional cubic FCC Al using ASE's bulk() function.

2. Replicate the cell 4 x 4 x 4. A conventional FCC cell contains 4 atoms, so the final structure should contain 256 atoms.

3. Write the structure to al_fcc_4x4x4.data using ASE's LAMMPS-data writer with atomic atom style.

Checkpoint: Do not continue until the script reports 256 atoms and the file al_fcc_4x4x4.data exists.

Part 2 - Prepare the five NPT simulations

Copy al_fcc_4x4x4.data and in.thermal_expansion to your SCC working directory. Your instructor will supply the Al interatomic potential and the SCC batch script.

In the LAMMPS input, replace TEMP1 TEMP2 TEMP3 TEMP4 TEMP5 with five temperatures in Kelvin. A useful range is roughly 100-900 K, with temperatures spread across the interval.

Setting	Value used	Why it matters
Ensemble	NPT	The cell volume must be free to change.
Pressure	0 bar	Approximates free thermal expansion with no imposed external pressure.
Timestep	1 fs	Resolves atomic vibrations for a metallic solid.
Equilibration	5 ps	Allows temperature and volume to relax toward equilibrium.
Production	10 ps	Provides a time window over which volume is averaged.
Runs	5 independent	Each temperature begins from the same original structure.

Why independent runs?

This is not a continuous heating experiment. For each temperature, LAMMPS reloads the same starting structure, assigns new velocities, equilibrates at the selected temperature and zero pressure, then collects production data. That makes each point in V(T) an independent equilibrium estimate rather than a point along a heating trajectory.

What LAMMPS writes

Each temperature produces a small file named thermal_<T>.dat. The file contains block-averaged timestep, temperature, pressure, and volume. You do not need a large dump trajectory for this activity.

Part 3 - Bring the results back to your laptop

After all five calculations finish, copy the five thermal_*.dat files to the same local folder as 02_analyze_thermal_expansion.py. Then run the analysis script locally.

What the analysis script does

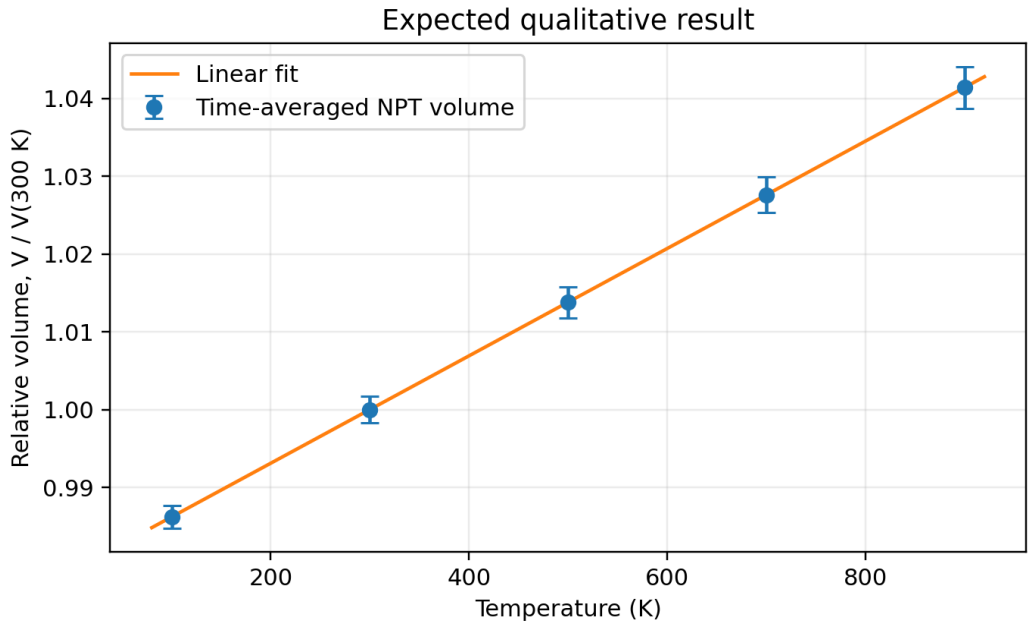
1. Finds all five thermal_.dat files automatically.
2. Calculates the mean simulated temperature, mean pressure, and mean volume for each run.
3. Uses the spread of the production block averages to show how much the equilibrium volume fluctuates.
4. Plots average simulation-cell volume as a function of target temperature.
5. Fits a straight line to V(T).
6. Calculates the volumetric and approximate linear thermal expansion coefficients.

From V(T) to thermal expansion

Quantity	Meaning
dV/dT	Slope of the fitted volume-versus-temperature line.
beta = (1/V)(dV/dT)	Volumetric thermal expansion coefficient.
alpha ~ beta/3	Approximate linear thermal expansion coefficient for an isotropic cubic solid such as FCC Al.

Important: The simulation directly gives you a change in volume, so beta is the quantity obtained most naturally. The commonly tabulated thermal expansion coefficient for Al is usually the linear coefficient alpha.

What should a successful result look like?



The figure above is schematic only. Your actual values depend on the interatomic potential and finite simulation time. The main expected trend is a positive, approximately linear increase in equilibrium volume with temperature over this range.

Expected physical behavior

Volume increases with temperature. Higher temperature produces larger-amplitude anharmonic vibrations, shifting the average atomic spacing outward.

Average pressure stays near zero. Instantaneous pressure can fluctuate strongly in a small MD cell, but its time average should remain near the imposed NPT target.

The five points are approximately linear. For this short classroom exercise, a linear V(T) fit is a useful first approximation even though real thermal expansion is temperature dependent.

The expansion coefficient is positive. For Al, the linear coefficient should be on the order of a few x 10^-5 K^-1, but the exact value depends on the supplied potential and the simulated temperature range.

What not to over-interpret

A 256-atom cell and tens of picoseconds are enough for a classroom demonstration, but they are not a convergence study. The exact coefficient can change with potential choice, system size, equilibration time, production length, and the temperature interval used for the fit. The goal here is to learn the workflow and the physics connecting ensemble choice to an observable property.

End-of-activity checks

☐ Did the ASE structure contain exactly 256 atoms?

☐ Did all five simulations start from the same original structure?

☐ Did you use NPT at zero pressure rather than NVT?

☐ Were the reported mean pressures reasonably centered around zero despite fluctuations?

☐ Did the average volume generally increase with temperature?

☐ Is the fitted thermal expansion coefficient positive and physically reasonable in order of magnitude?

Think about these before you leave

• Why would NVT be unable to measure thermal expansion directly?

• Why is a time-averaged volume more meaningful than a single instantaneous volume?

• Why might the pressure fluctuate much more strongly than the volume?

• What would you change if you wanted a publication-quality thermal expansion coefficient rather than a classroom estimate?

Take-home idea: Molecular dynamics is not only a way to watch atoms move. When the ensemble and analysis are chosen correctly, atomic trajectories can be converted into macroscopic materials properties that can be compared directly with experiment.